**MCCG11303/MECG11303/MEME19703/UECS3183**

**Data Visualisation Techniques**

**Assignment 2 (20%)**

## Instructions

1. This assignment is an individual assessment.
2. You are required to submit your solution by **10:00 PM, 4 September 2025 (Friday)**.
3. You solution consists of three items:
   - A ready-to-run Python file.
   - A raw data file in CSV format.
   - A presentation video in MP4 format.

   Please zip your files and name the zip file using the following file name for submission:

   [Course Code]_[Programme Abbreviation]_[Your Student ID]

   For example, UECS3183_SE_2500123.
4. Please make sure you submit the correct, complete and final version of your solution. The solution file is to be submitted to the given **Microsoft Teams** link.

## WARNING OF PLAGIARISM

5. Your solution shall be checked for the originality. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have committed plagiarism in the submitted solution. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

**Task**

1. Gather and prepare your own data from any appropriate data sources. Your data should preferably have at least 10,000 records and 8 fields of both numerical and categorical values, including at least one date field. You can use your data from Assignment 1.

2. Perform all the necessary data cleaning.

3. You are to create an **interactive dashboard** to tell a story about your dataset (objective of the dashboard). You are to choose the story that you want to tell.

4. Using **Streamlit** or **Jupyter Notebook with Ipywidgets** or any appropriate combination of libraries, build an interactive dashboard with Python to tell the story of your data. Include your name, ID, course code, and program as comments in your source code. Also include instructions on how to generate the dashboard as comments in your source code.

5. Create a short presentation (8 to 15 minutes) that
   - describe the data sources,
   - describe the story (objective) of the visualization,
   - demonstrate the interactive widgets of your dashboard, and
   - discuss any insights from the data (with justification from your dashboard).

   You can record your presentation using MS Teams or any appropriate software.

**Marks allocation (20 marks in total)**

- Dashboard ( 10 marks)
- Presentation (10 marks)

See marking rubric for details.

**Marking Rubric**

| | Poor (1 – 4 marks) | Average (5 – 8 marks) | Good (9 – 10 marks) |
|---|---|---|---|
| **Presentation (10 marks)** | • Fail to provide justification onthe data.<br>• Has not enough contents.<br>• Fail to demonstrate the dashboard.<br>• Fail to provide insightfulinformation. | • Able to provide limited justification on the data.<br>• Has sufficient contents.<br>• Able to demonstrate the dashboard.<br>• Able to provide some insightful information. | • Able to provide strong justification on the data.<br>• Has sufficient and interestingcontents.<br>• Able to demonstrate the dashboard effectively.<br>• Able to provide interesting and insightful information. |
| **Dashboard (10 marks)** | • Fail to build or build less than 4 charts.<br>• Charts are barely functional.<br>• Charts provide limited analysis on the data.<br>• The dashboard is not user-friendly and not interactive. | • Able to build at least 4 charts.<br>• Charts are functional with a good layout.<br>• Charts provide sufficient analysis on the data.<br>• The dashboard is interactive with at least 2 widgets. | • Able to build at least 4 charts and at least one advanced chart.<br>• Charts illuminate every aspect of the story and guides the user'sattention.<br>• Charts provide advanced analysis and insight on the data.<br>• The dashboard is interactive with at least 4 widgets, is user-friendly, and is visually appealing with nice styling. |